

STRUCTURAL AND COMPUTATIONAL RESULTS ON THE FINITE-SUPPORT HIGHWAY CONJECTURE FOR LANGTON’S ANT

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ABSTRACT. Langton’s ant is a deterministic reversible lattice walker on $\mathbb{Z}^2$ whose long-term behaviour, for every finite initial colouring, is empirically the formation of a period-104 “highway”. Whether this holds for *every* finite initial black support is a well-known open problem. This paper collects, in a uniform and rigorous framework, a body of structural and computational results on this conjecture. We prove: an exact, decidable necessary-and-sufficient criterion for an infinite periodic turn word to be the trace of some finite-support colouring (Theorem 3.1); a heading/black-count mod-four invariant and a reduction of the two-dimensional orbit to a one-dimensional black-count walk; an exact conjugacy of the ant with edge-toggling on a Tait (medial) graph, together with a cycle-rank surgery identity; and two independent proofs that *no finite-support clean translator with zero net growth exists*—one homological (the *even-winding theorem*) and one via a *signed mod-four wake-residue* identity, which further yields the strand-density bound $g \geq 2\|d\|_\infty$ for every finite-support repeating trace of nonzero drift. Consequently every finite-support periodic highway has net black growth a positive multiple of four. For diagonal drift we prove extremal-line rigidity and exclude transverse width two at every period. A computer-assisted crossing-sequence theorem then excludes width four at every period: two independent exact enumerators produce the same twelve-edge blank-column graph, and Lean checks that every edge lowers rank. Thus a diagonal periodic highway has transverse width at least six. On the computational side we give an exact finite black seed for the standard highway and exclude by a rank-audited exhaustive search—using no consequence of the residue identity, and applying the exact criterion to every positive-growth nonzero-drift leaf—every finite-support periodic highway of nonzero drift and period at most 48. We isolate two logically separate remaining problems: a universal entrance/bounded-retreat problem and the all-period classification of positive-growth periodic traces. We describe a Lean 4 formalization of selected algebraic kernels. No claim of a proof or disproof of the highway conjecture is made.

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1. INTRODUCTION

1.1. The automaton and the conjecture. Langton’s ant moves on the square lattice $\mathbb{Z}^2$, each cell coloured *white* (0) or *black* (1). A pointer (the *ant*) occupies a cell and faces one of the four cardinal directions. One step of the dynamics is:

if the current cell is white, turn right (90° clockwise); if it is black, turn left; then flip the colour of the current cell and move one unit forward in the new heading.

We always assume the *finite-support* condition: the set of initially black cells is finite. Translating and rotating, we normalise the initial ant to be at the origin facing north.

Iterating from the all-white configuration produces, after a transient of 9977 steps, a self-similar diagonal structure—the *highway*—that repeats with least period 104 and displaces the ant by $(\pm 2, \pm 2)$ per period. Extensive experiment suggests the same eventual behaviour for every finite initial colouring.

Conjecture 1.1 (Finite-support highway conjecture). For every finite initial black set $B_0 \subset \mathbb{Z}^2$ and every initial pose, there is a time after which the ant’s future turn trace and path agree with the standard period-104 highway up to lattice translation, quarter-turn rotation, and cyclic phase shift. Finite cells outside the future path may remain as stationary debris; equality of the entire colouring is not asserted.

Conjecture 1.1 is stronger than *unboundedness* of the trajectory (that the ant visits infinitely many cells), which is a theorem of Bunimovich and Troubetzkoy [1]. The ant is computationally universal, and associated prediction problems are undecidable, on suitable infinite backgrounds [5]; we draw no inference from this about the finite-support conjecture beyond the evident point that no finite cutoff exhausts an infinite range. Gajardo [3] gives an algebraic characterisation of the forbidden factors of the ant’s trace subshift. Recent work on *generalised* ants exhibits rules with infinitely many highways [4] and rules with non-highway emergent behaviour [6]; quantitative confinement bounds for the classical ant are studied in [2]. Conjecture 1.1 remains open.

1.2. What is proved here, and what is not. This paper is deliberately explicit about status. We prove a number of exact structural theorems and finite computational theorems; *none* of them resolves Conjecture 1.1. Section 11 states the two logically separate unresolved problems. Every statement below is labelled as an exact theorem, a finite computational theorem, a conjecture, or an explicitly open problem. The main contributions are:

- (1) An exact periodic-trace criterion (Theorem 3.1) turning “is w^ω a finite-support highway?” into a finite computation, with an explicit seed.
- (2) The heading/black-count invariant and the black-count path reduction (Section 2.3).
- (3) The Tait-graph / boundary-loop conjugacy and the cycle-rank surgery identity (Section 4).
- (4) The even-winding theorem: no zero-growth clean translator (Theorem 6.1), hence growth $\in 4\mathbb{Z}_{>0}$ for every highway.
- (5) The signed mod-four wake-residue theorem (Theorem 7.1) and the strand-density bound $g \geq 2 \max(|a|, |b|)$.
- (6) Residue-free, rank-audited exclusion of all finite-support periodic highways of period ≤ 48 (Section 10).
- (7) A two-endpoint collision-chain parity theorem relevant to the entrance problem (Section 8).
- (8) Extremal-line rigidity and all-period exclusions of diagonal transverse widths two and four; the width-four result has an explicit computer-assisted local classification boundary (Section 9).
- (9) A Lean 4 formalization of selected algebraic kernels (Appendix A).

The presentation is organised so that failed and merely heuristic arguments are recorded honestly (Section 11), because they delimit the true difficulty of the problem.

2. CONVENTIONS AND ELEMENTARY INVARIANTS

2.1. State, step, trace. A *state* is a triple $s = (B, p, q)$ with $B \subset \mathbb{Z}^2$ finite (the black set), $p \in \mathbb{Z}^2$ (the ant position) and $q \in \mathbb{Z}/4\mathbb{Z}$ the heading, numbered $0 = N, 1 = E, 2 = S, 3 = W$. Let $u_0 = (0, 1), u_1 = (1, 0), u_2 = (0, -1), u_3 = (-1, 0)$ be the unit heading vectors. The *turn at s* is R (right) if $p \notin B$ and L (left) if $p \in B$; write $\varepsilon = +1$ for R and $\varepsilon = -1$ for L. One step is

$$s \mapsto (B \triangle \{p\}, p + u_{q+\varepsilon}, q + \varepsilon),$$

where $\triangle$ is symmetric difference. The *trace* of an orbit is the infinite word $w \in \{R, L\}^{\mathbb{N}}$ of turns encountered.

Definition 2.1 (Periodic terminology). Let $w = w_0 \cdots w_{N-1}$ be a nonempty turn word and follow its induced lattice path from a specified initial heading. The word is *heading-resetting* if the heading after N turns equals the initial heading. In that case its *drift* is $p_N - p_0$, and its *growth* is $\#\{i : w_i = R\} - \#\{i : w_i = L\}$. An orbit has an *eventually periodic turn trace* if, from some time t_0 , its turns are w^ω . A finite-support tail of this form with w heading-resetting is called a *repeating trace*; if its drift is nonzero, it is a *periodic highway*. Its *least trace period* is the smallest possible N . Highways are compared up to lattice translation, quarter-turn rotation, and cyclic phase shift of w .

The dynamics is reversible: the pre-image of (B, p, q) is obtained by stepping backward one cell, recolouring, and undoing the turn; hence orbits never merge.

2.2. HV partition and alternating visits.

Proposition 2.2 (P1: HV/checkerboard partition). *Along any orbit the heading parity $q_t \bmod 2$ and the checkerboard parity $(p_{t,x} + p_{t,y}) \bmod 2$ each change at every step, so their difference is a constant of the motion. Consequently each cell z is entered by the ant always horizontally or always vertically; the admissible directed lattice edges have a fixed orientation, and each cell has exactly two incoming and two outgoing directed edges.*

Proof. Each step turns by ± 1 quarter turn, so $q_{t+1} - q_t$ is odd and $q_t \bmod 2$ alternates. The displacement $u_{q_{t+1}}$ has odd L^1 -length, so $(p_{t,x} + p_{t,y}) \bmod 2$ also alternates. Their sum is therefore invariant. Fix a cell z ; whenever $p_t = z$ the value of $(p_{t,x} + p_{t,y}) \bmod 2$ is fixed, hence so is $q_t \bmod 2$: the arrival is always via a horizontal edge or always via a vertical edge. A vertical-arrival cell has its two vertical edges incoming and horizontal edges outgoing, and vice versa. $\square$

Call a cell *V-type* if the ant always arrives vertically and *H-type* otherwise. We write $\chi(x, y) = (-1)^{x+y}$ for the checkerboard sign; V/H type is determined by χ together with the fixed invariant of Proposition 2.2.

Proposition 2.3 (P2: alternating visit rule). *At any fixed cell the turns encountered alternate strictly: R, L, R, L, ... or L, R, L, Since a cell is white iff it will next cause an R, and only finitely many cells are initially black, all but finitely many cells have R as their first encountered turn.*

Proof. Each visit flips the colour, so the colour seen at a cell alternates; the turn is a function of colour, hence alternates too. A cell that is white at its first visit begins with R; only cells in B_0 can begin with L. $\square$

Theorem 2.4 (Unboundedness [1]). *Every finite-support orbit visits infinitely many cells.*

Proof sketch. If the visited set were finite, determinism and reversibility would make the orbit periodic in the full state. Take a visited cell z maximising $x + y$; its north and east neighbours are unvisited. If z is entered horizontally, it can only be entered from the west. During a full state period its colour returns, so its alternating visits include both a white/R and a black/L visit; the corresponding departures include the unvisited north neighbour. If z is entered vertically, it can

only be entered from the south, and the two colours force departures including the unvisited east neighbour. Either case is a contradiction. $\square$

2.3. The heading/black-count invariant and path reduction. Let B_t, p_t, q_t be the state before step t , $b_t = |B_t|$, and $r_t = b_{t+1} - b_t \in \{\pm 1\}$.

Proposition 2.5 (**P4, P9**: heading/count invariant and path reduction). *For all t ,*

$$q_t - b_t \equiv q_0 - b_0 \pmod{4}, \quad (1)$$

and the two-dimensional path is a function of the one-dimensional black-count walk:

$$p_{t+1} - p_t = u_{q_0 - b_0 + b_{t+1}}. \quad (2)$$

In particular a heading-resetting period changes b by a multiple of 4.

Proof. A white/R step has $r_t = +1$ and $q_{t+1} - q_t = +1$; a black/L step has both differences -1 . Thus $q_t - b_t$ is invariant, giving (1). The move occurs after the turn, so $q_{t+1} = q_0 - b_0 + b_{t+1} \pmod{4}$ and (2) follows. $\square$

Equation (2) does not say every nonnegative nearest-neighbour walk is colour-realizable: repeated visits still must request alternating turns (Proposition 2.3). Differentiating the centred first moment $J_t = \sum_{z \in B_t} (z - p_t)$ gives the exact identity

$$J_{t+1} - J_t = -b_{t+1} u_{q_0 - b_0 + b_{t+1}}, \quad (3)$$

which is invariant under simultaneous translation of the black set and the ant (**P10**); it will quantify the wake of a growing highway.

3. THE EXACT PERIODIC-TRACE CRITERION

The following theorem converts the search for a nonstandard highway into a finite, exact decision problem. Fix an initial heading q_0 and a proposed finite turn word $w = w_0 \cdots w_{N-1}$. Compute the associated path $p_0, p_1, \dots, p_N$ purely from w (Proposition 2.2 makes each step well defined). Suppose the heading is restored ($q_N = q_0$) and the displacement

$$d = p_N - p_0 \neq 0.$$

Define an equivalence on phases by $i \sim j \iff p_i - p_j \in \mathbb{Z}d$. For each class I pick a representative b_I and integers a_i with $p_i = b_I + a_i d$ ($i \in I$). For $n \geq \min_{i \in I} a_i$, let $S_{I,n}$ be the word formed by all w_i ($i \in I$) with $a_i \leq n$, ordered by the pair $(n - a_i, i)$.

Theorem 3.1 (**P3**: finite-support periodic-trace criterion). *The infinite word w^ω is the trace of Langton's ant from some finite-support colouring if and only if, for every class I :*

- (i) *every $S_{I,n}$ with $\min_I a_i \leq n \leq \max_I a_i$ alternates strictly between R and L; and*
- (ii) *the stabilised word $S_{I, \max_I a_i}$ begins with R.*

When these hold, an explicit finite black seed is

$$B = \{ b_I + nd : \min_I a_i \leq n < \max_I a_i \text{ and } S_{I,n} \text{ begins with L} \},$$

the union over all classes I ; the two bounds make B finite and $S_{I,n}$ well defined.

Proof. $S_{I,n}$ is precisely the chronological turn sequence at the physical cell $b_I + nd$ over the infinite trajectory w^ω : the occurrence of phase $i \in I$ in period $k \geq 0$ is at $p_i + kd = b_I + (a_i + k)d$, which equals $b_I + nd$ exactly when $k = n - a_i \geq 0$, and chronological order is the lexicographic order of $(n - a_i, i)$.

Necessity. Each visit flips the cell, so its turn sequence alternates: (i) is necessary. Once $n \geq \max_I a_i$ every phase of I is present and the ordering no longer depends on n ; the sequence has stabilised, and infinitely many cells share this first turn. By Proposition 2.3 only finitely many cells may start with L, so the stable first turn is R: (ii) is necessary.

Sufficiency. Colour exactly B black. At each boundary cell the chosen colour reproduces the first symbol of $S_{I,n}$, and strict alternation supplies every later symbol because the cell flips at each visit. At every stabilised cell (ii) makes the first encounter R, matching its white initial colour. Induction on chronological time shows the ant follows w^ω forever; the position is displaced by $d \neq 0$ every N steps, so this is a highway. $\square$

Remark 3.2. There is an equivalent one-sequence test: in each class sort all entries by decreasing a_i , breaking ties by increasing i ; every $S_{I,n}$ is a suffix of this stabilised word obtained by deleting complete high-level blocks, so it suffices to check that the single stabilised word alternates and begins with R. To index the classes of $\mathbb{Z}^2/\mathbb{Z}d$ exactly, write $g = \gcd(a, b)$ and $d = g(a', b')$ with $\gcd(a', b') = 1$, and choose integers r, s with $ra' + sb' = 1$. Then

$$(b'x - a'y, (rx + sy) \bmod g)$$

is a complete invariant of the coset of (x, y) . Indeed, the matrix $\begin{pmatrix} b' & -a' \\ r & s \end{pmatrix}$ is unimodular, and in its coordinates translation by d becomes translation by $(0, g)$. This is the checker used in all computations of Section 10.

4. THE TAIT-GRAPH / BOUNDARY-LOOP CONJUGACY

Normalise the HV partition (Proposition 2.2) so that even cells receive vertical arrivals. In the *frozen* all-white routing—the permutation on directed states that ignores colour flips—the admissible directed arcs form clockwise directed four-cycles around one checkerboard class of unit plaquettes. Take those plaquettes, labelled by their lower-left corners, as the vertices of a rotated square lattice. Each ant cell $z = (x, y)$ is the crossing of exactly two white four-cycles and corresponds to the Tait edge

$$e_z = \begin{cases} \{(x, y-1), (x-1, y)\}, & x+y \text{ even,} \\ \{(x-1, y-1), (x, y)\}, & x+y \text{ odd.} \end{cases} \quad (4)$$

Let $H_B = \{e_z : z \in B\}$ be the finite edge subgraph of black cells.

Theorem 4.1 (P5: boundary-loop conjugacy). *White and black are the two smoothings of the corresponding directed crossing. Consequently:*

- (a) *the frozen routing loops are exactly the boundary components of a regular neighbourhood of H_B , together with the untouched four-cycles;*
- (b) *the ant traverses its current boundary loop and then toggles e_z ; and*
- (c) *every step either merges two frozen loops into one or splits one loop into two.*

This is an exact conjugacy, not a heuristic.

Proof. Frozen (colour-ignoring) routing is the permutation of directed edges sending each edge to the next one the ant would follow under a fixed local turn rule. At each cell z , right and left turns are precisely the two smoothings of the associated 4-valent medial crossing, and the Tait edge e_z records which diagonal pair of plaquette corners the crossing joins.

Freeze a colouring B . Away from H_B every crossing has the white smoothing. Those smoothings assemble into the boundary circuits of a regular neighbourhood $\mathcal{N}$ of H_B : occupied edges are thickened to bands and incident Tait vertices to disks, while untouched white plaquettes contribute trivial four-cycles. Thus the frozen loops are the components of $\partial\mathcal{N}$ together with those trivial cycles. The live ant advances on its current boundary loop and then toggles z , which adds or deletes the single band e_z . This surgery reconnects two local boundary arcs: it splits a component if both arcs belonged to the same loop and merges them if they belonged to different loops. These are the only cases. $\square$

Let $E = |B|$ be the number of black (occupied) Tait edges, V the number of incident Tait vertices, k the number of nonempty connected components, and $\beta = E - V + k$ the cycle rank.

Theorem 4.2 (P6: cycle-rank surgery identity). *A regular neighbourhood of H_B has $k + \beta = E - V + 2k$ boundary components. Relative to the V separate all-white plaquette loops, the change in loop count is*

$$\Delta C = E - 2V + 2k, \quad E + \Delta C = 2\beta. \quad (5)$$

Adding a bridge merges two loops and preserves β ; adding a cycle edge splits a loop and raises β by one. Removing a bridge splits a loop and preserves β ; removing a nonbridge merges two loops and lowers β by one.

Proof. A regular neighbourhood of a plane graph with V vertices, E edges and k components is a compact planar surface with k genus-zero pieces and Euler characteristic $V - E$. If it has b boundary circles, then $V - E = 2k - b$, so $b = E - V + 2k = k + \beta$, where $\beta = E - V + k$ is the first Betti number. Formula (5) is the difference from the V trivial plaquette loops. The surgery cases are the standard effect of adding/deleting a bridge (does not change β ; changes the loop count by ∓ 1 through merge/split) versus a cycle edge (changes β by ± 1 ; opposite merge/split), read off from Theorem 4.1(c). $\square$

Proposition 4.3 (P7: eventual record edges are bridges). *Once a new bounding-box record lies strictly beyond the bounding box of the finite initial support, its first visit is white, and a new east, north, west, or south record turns south, east, north, or west, respectively. Its Tait edge has a previously untouched endpoint, so it is a bridge. Hence sufficiently advanced record creation cannot increase β : any unbounded production of cycle rank occurs strictly inside already-explored territory.*

Proposition 4.4 (No monotone affine invariant). *No nonconstant affine combination of (E, V, k, β) is monotone along the blank orbit. In particular, each of E , V , k , and β separately fails to be monotone. Thus no affine combination of these four graph statistics can serve as a monotone or Lyapunov potential along even the blank orbit.*

Proof. Since $\beta = E - V + k$ identically, $aE + bV + ck + d\beta + e = (a + d)E + (b - d)V + (c + d)k + e$: the affine combinations of (E, V, k, β) are exactly those of (E, V, k) , nonconstant precisely when $\lambda := (a + d, b - d, c + d) \neq 0$. One step flips one cell, hence inserts or deletes exactly one Tait edge $e_z = \{u, v\}$; insertion has exactly four cases, and deletion negates each:

insertion of $e_z = \{u, v\}$	$(\Delta E, \Delta V, \Delta k)$	$\Delta \beta$
u, v both new	$(1, 2, 1)$	0
exactly one of u, v new	$(1, 1, 0)$	0
u, v both old, distinct components	$(1, 0, -1)$	0
u, v both old, same component	$(1, 0, 0)$	1

So every increment lies in $S_0 = \pm\{(1, 2, 1), (1, 1, 0), (1, 0, -1), (1, 0, 0)\}$, and β rises exactly when the inserted edge closes a cycle. All eight members of S_0 occur on the blank orbit within 12,000 steps. S_0 is centrally symmetric by construction and contains the independent $(1, 1, 0), (1, 0, -1), (1, 0, 0)$, so it spans $\mathbb{R}^3$. Given $\lambda \neq 0$, spanning yields $v \in S_0$ with $\lambda \cdot v > 0$ (replace v by $-v$ if needed), while $\lambda \cdot (-v) < 0$ and $-v$ is realised too: λ rises at one step and falls at another. Equivalently 0 is interior to $\text{conv } S_0$. See `verify_nomonotone.py`. $\square$

5. CHARGES AND CLEAN-TRANSLATOR RESTRICTIONS

Definition 5.1. A *clean finite translator with nonzero drift* is a finite state such that for some $N > 0$ and $d \in \mathbb{Z}^2 \setminus \{0\}$,

$$B_N = B_0 + d, \quad p_N = p_0 + d, \quad q_N = q_0.$$

“Clean” means the complete black set translates: no stationary debris, no newly emitted wake. This is *not* the ordinary highway, which grows a wake.

5.1. Additive $\mathbb{F}_2$ charges and axis alignment. Work in the Laurent ring $\mathbb{F}_2[X^{\pm 1}, Y^{\pm 1}]$. Let $V_t = 1$ if q_t is vertical, $H_t = 1 - V_t$, and

$$A_t = \sum_{(x,y) \in B_t} X^x + V_t X^{x_t}, \quad D_t = \sum_{(x,y) \in B_t} Y^y + H_t Y^{y_t}.$$

Proposition 5.2 (P11, Step 4). *A_t and D_t are conserved exactly. Hence a clean translator with $d = (a, b)$ satisfies $A = X^a A$ and $D = Y^b D$ in the integral domain $\mathbb{F}_2[X^{\pm 1}, Y^{\pm 1}]$; since $A(1) = |B| + V$ and $D(1) = |B| + H$ cannot both vanish ($V + H = 1$), a and b cannot both be nonzero. Every clean translator is axis-aligned.*

Proof. For A_t : if the incoming heading is vertical the cell toggle at column x_t cancels the disappearing vertical correction; if horizontal, the post-turn motion is vertical so $x_{t+1} = x_t$ and the toggle cancels the newly appearing correction. Conservation and translation covariance give $A = X^a A$. In an integral domain $a \neq 0$ forces $A = 0$, similarly $b \neq 0$ forces $D = 0$; but $A(1) + D(1) = 2|B| + 1 = 1 \neq 0$ in $\mathbb{F}_2$. The row statement is the rotated argument. $\square$

Proposition 5.3 (additive-charge cocycle). *Let A be an abelian group and $w : \mathbb{Z}^2 \rightarrow A$. If there exist heading corrections $F_q : \mathbb{Z}^2 \rightarrow A$ such that $\sum_{z \in B} w(z) + F_q(p)$ is conserved under*

$$F_{q+1}(z + u_{q+1}) - F_q(z) = -w(z) \quad (\text{R}), \quad F_{q-1}(z + u_{q-1}) - F_q(z) = +w(z) \quad (\text{L}), \quad (6)$$

then w satisfies the four-corner equation

$$w(x+1, y+1) + w(x+1, y) + w(x, y+1) + w(x, y) = 0 \quad (7)$$

and $4w = 0$. Conversely, for $A = \mathbb{Z}/4\mathbb{Z}$, the four-corner equation yields such heading corrections. Over $\mathbb{F}_2$ its solutions are exactly $w(x, y) = \alpha(x) + \beta(y)$.

Proof. For necessity, eliminate the four F_q from (6) around a unit plaquette to obtain (7). At one cell z , the R-at-S/L-at-N relations give $F_S(z) = F_N(z) + 2w(z)$, while the L-at-S/R-at-N relations give $F_S(z) = F_N(z) - 2w(z)$; hence $4w(z) = 0$.

For sufficiency over $\mathbb{Z}/4\mathbb{Z}$, compose pairs of equations in (6) and define F_N by the diagonal increments

$$\begin{aligned} F_N(x+1, y+1) - F_N(x, y) &= -w(x, y) + w(x+1, y), \\ F_N(x+1, y-1) - F_N(x, y) &= w(x, y-1) - w(x+1, y-1). \end{aligned}$$

The circulation around each elementary diamond is a sum of two four-corner expressions plus a multiple of $4w$, so it vanishes. Thus these increments integrate consistently on both parity sublattices. Set $F_S = F_N - 2w$ and define F_E, F_W from F_N using (6); substitution verifies all eight local relations, with the only ambiguity equal to $4w = 0$. Finally, over $\mathbb{F}_2$, taking first differences of (7) shows the horizontal difference depends only on x (equivalently the vertical difference only on y), and integration gives exactly $w(x, y) = \alpha(x) + \beta(y)$. $\square$

5.2. The strand decomposition. Let $g = \#R - \#L$ be the net growth of one period of a repeating word. Restoring the heading forces $g \equiv 0 \pmod{4}$ (Proposition 2.5), and $g \geq 0$ because g equals the number of *odd* stabilised translation classes (Theorem 3.1).

Theorem 5.4 (P12: canonical endpoint decomposition). *Every repeated trace has a canonical decomposition of its one-period signed toggle pattern $D(z) = \#\{R \text{ at } z\} - \#\{L \text{ at } z\}$:*

$$D = S + (T_d - 1)A, \quad S = \sum_{i=1}^g \delta_{c_i}, \quad (8)$$

where $(T_d A)(z) = A(z - d)$, A is a finite moving widget, and the c_i are the bases of the g growing strands (one +1 base per odd stabilised class).

Proof. D is finitely supported. Partition its support into translation classes under $z \mapsto z + d$. In one class choose a base c and put $f(n) = D(c + nd)$, a finitely supported function on $\mathbb{Z}$. Let $s = \sum_n f(n)$ and $h = f - s\delta_0$, so $\sum_n h(n) = 0$. With $(Ta)(n) = a(n-1)$, define $a(n) = \sum_{m>n} h(m)$. Then a has finite support and $a(n-1) - a(n) = h(n)$, hence $f = s\delta_0 + (T-1)a$. Summing these classwise decompositions gives (8) with finite A .

It remains to identify s . Within a translation class, the alternating-visit rule makes the one-period net toggle equal to $+1$ when the stabilised class word has odd length and 0 when it has even length. Thus $s \in \{0, 1\}$, and S is a sum of one unit mass for each odd class. The number of odd classes is the net black growth $g = \#R - \#L$, proving the stated form. $\square$

Corollary 5.5. *If $g = 0$, a finite stationary set can be deleted after finitely many periods to leave a clean finite translator with drift d .*

Proof. Here $S = 0$, so $D = (T_d - 1)A$. Let b_n be the indicator of the black set at the n th period boundary. Translation covariance and telescoping give

$$b_n = b_0 + \sum_{k=0}^{n-1} T_{kd}D = b_0 + (T_{nd} - 1)A = R + T_{nd}A, \quad R := b_0 - A.$$

Both R and A are finitely supported, so their supports are disjoint for all large n ; because b_n is $\{0, 1\}$ -valued, each is then the indicator of a finite set. A nonzero-drift periodic path visits every fixed cell only finitely often. At a sufficiently late boundary the ant therefore never revisits R ; deleting it changes no future turn and leaves the finite pattern $T_{nd}A$, which translates exactly by d each period. $\square$

Sections 6–7 prove that the resulting clean translator is impossible.

6. THE EVEN-WINDING THEOREM

Theorem 6.1 (P15: no zero-growth clean translator). *There is no finite black pattern and ant state that evolves in finitely many steps to exactly one nonzero translate of itself with restored heading and zero net black growth. Equivalently, no clean finite translator with nonzero drift exists.*

Proof. Suppose $(A, p, q) \mapsto (A + d, p + d, q)$ in one word of length N with A finite and $d \neq 0$. By Proposition 5.2 the drift is axis-aligned. Since a clean translator has zero net growth, $N = 2\#R$ is even. Each step changes $x + y$ modulo two, so the nonzero drift coordinate is even. After a quarter-turn rotation write $d = (L, 0)$ with $L > 0$ even.

Quotient and the visit count. Reversibility and translation covariance give, for all $n \in \mathbb{Z}$ and phases j , $A_{nN+j} = A_j + nd$ and $p_{nN+j} = p_j + nd$. Quotient the full bi-infinite trace by translation through d to obtain a closed walk on the cylinder $C_L = (\mathbb{Z}/L\mathbb{Z}) \times \mathbb{Z}$; the checkerboard HV type descends because L is even. Fix a quotient vertex v with physical representative z . Each phase j whose position lies over v has a unique level m_j with $p_j = z + m_j d$, and phase j in period $-m_j$ visits z ; conversely every visit to z arises once this way. Thus the visits to v in one quotient period are exactly the visits to z over the bi-infinite translated orbit. Because at each fixed phase the black pattern is a finite translate, z is white far in both time directions; its turns alternate, begin with R, end with L, and are equally many of each. Hence every quotient vertex v has

$$n_v = 2k_v, \quad k_v \text{ R's and } k_v \text{ L's}, \quad (9)$$

with no footprint or gap assumption ($k_v = 0$ allowed).

Edge multiplicities. For each undirected cylinder edge let its multiplicity be the number of traversals by the projected one-period trace. Proposition 2.2 orients each admissible edge uniquely, so at a vertex one pair of incident edges carries all arrivals and the other all departures:

$$H_{v,-} + H_{v,+} = V_{v,-} + V_{v,+} = 2k_v, \quad (10)$$

where $H_{v,\pm}$ are the west/east horizontal and $V_{v,\pm}$ the south/north vertical multiplicities. Reducing the horizontal equality mod 2 shows consecutive horizontal edges share parity, so each row y has one horizontal parity $h_y \in \mathbb{F}_2$. The same propagation up a column, together with finite vertical support (edges far below/above the trace have multiplicity 0), forces *every vertical multiplicity to be even*: $V_e = 2G_e$, $G_e \in \mathbb{Z}_{\geq 0}$.

Local parity and the telescope. A local HV parity computation gives, at both cell types, $k_v \equiv h_y \pmod{2}$: at a vertical-arrival cell the horizontal output has parity h_y and the vertical input is even; at a horizontal-arrival cell the vertical output is even and the horizontal input has parity h_y . Dividing the vertical part of (10) by two gives the exact identity $k_{x,y} = G_{x,y-1/2} + G_{x,y+1/2}$. Fix a column x and sum over all rows:

$$\sum_y h_y \equiv \sum_y (G_{x,y-1/2} + G_{x,y+1/2}) \equiv 0 \pmod{2}, \quad (11)$$

because every interior half-edge G occurs twice and the two exterior terms vanish by finite support.

Homological contradiction. Cut the cylinder along a vertical seam between two adjacent columns. The algebraic winding number w of the projected closed trace is the signed number of seam crossings; there is one horizontal seam edge per row, of parity h_y , and signs disappear mod 2, so by (11)

$$w \equiv \sum_y h_y \equiv 0 \pmod{2}.$$

But the lift of the closed trace runs from p to $p + (L, 0)$ and therefore winds exactly once, $w = 1$. This contradiction proves the theorem for drift $(L, 0)$; quarter-turn rotations, including a half-turn, cover the remaining axis-aligned drifts. $\square$

The telescoping step (11) is exactly the finite algebraic core machine-checked in Lean (Appendix A, `evenWindingParityCore`).

Corollary 6.2 (growth is a positive multiple of four). *Every finite-support periodic highway has net black growth*

$$g \in 4\mathbb{Z}_{>0}. \quad (12)$$

Proof. Zero drift confines the ant to finitely many positions of one period, excluded by unboundedness (Theorem 2.4). Negative growth cannot repeat forever since $b_t \geq 0$. A heading-resetting period has $g \equiv 0 \pmod{4}$ (Proposition 2.5). By Corollary 5.5 a $g = 0$ repeating trace with nonzero drift separates into stationary debris and a clean moving packet, contradicting Theorem 6.1. Hence $g > 0$ and $g \in 4\mathbb{Z}_{>0}$. $\square$

The ordinary highway has $g = 12$, consistent with (12).

7. THE SIGNED MOD-FOUR WAKE-RESIDUE THEOREM

We now give an independent proof of zero-growth exclusion that also produces a strand-density bound. Recall $\chi(x, y) = (-1)^{x+y}$ and the strand bases $c_1, \dots, c_g$ of Theorem 5.4.

Theorem 7.1 (P22: signed mod-four wake-residue identity). *Let a finite-support repeating trace have restored heading, nonzero drift $d = (a, b)$, growth g , and canonical decomposition (8). Then:*

(i) *if $a \neq 0$, then for every residue $r \in \mathbb{Z}/|a|\mathbb{Z}$,*

$$\sum_{i: c_{i,x} \equiv r \pmod{|a|}} \chi(c_i) \equiv 2 \pmod{4}; \quad (13)$$

(ii) *if $b \neq 0$, then for every residue $s \in \mathbb{Z}/|b|\mathbb{Z}$,* $\sum_{i: c_{i,y} \equiv s \pmod{|b|}} \chi(c_i) \equiv 2 \pmod{4};$

(iii) *if $a = 0$ the signed sum over any exact column is $\equiv 0 \pmod{4}$, and if $b = 0$ over any exact row.*

Proof. Fix $a \neq 0$ and an arbitrary $\alpha: \mathbb{Z}/|a|\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$. Put the black-cell weight

$$w(x, y) = \chi(x, y) \alpha(x) \in \mathbb{Z}/4\mathbb{Z}. \quad (14)$$

Then $4w = 0$ and w satisfies the four-corner equation (7) because χ changes sign on each unit step; by Proposition 5.3 there are heading potentials F_q with, on an R step $F_{q+1}(z + u_{q+1}) - F_q(z) = -w(z)$ (a black cell of weight w is added) and on an L step $F_{q-1}(z + u_{q-1}) - F_q(z) = +w(z)$.

Composing the north-to-east R equation with the east-to-north L equation gives $F_N(z + (1, 1)) - F_N(z) = w(z + (1, 0)) - w(z) = -\chi(z)(\alpha(x) + \alpha(x + 1))$, and via $F_S = F_N - 2w$ the same increment along $(1, -1)$; since the endpoints differ by $(0, 2)$, F_N is vertically two-periodic. Restored heading gives $g \equiv 0 \pmod{4}$, so the period is even and $a + b$ is even; hence translation by d preserves χ and $w(z + d) = w(z)$. Telescoping one complete horizontal residue cycle and using vertical two-periodicity yields the monodromy

$$F_q(z + d) - F_q(z) \equiv 2 \sum_{r \in \mathbb{Z}/|a|\mathbb{Z}} \alpha(r) \pmod{4} \quad (15)$$

for all four headings q and either sign of a (the telescoped total is $-2\chi(z) \sum_r \alpha(r)$, and $-2 \equiv 2 \pmod{4}$).

Charge conservation over one period reads

$$\sum_z D(z) w(z) + F_q(p + d) - F_q(p) = 0. \quad (16)$$

Because w is d -periodic and $(T_d A)(z) = A(z - d)$, the widget terms cancel: $\sum_z (T_d A)(z) w(z) = \sum_z A(z) w(z)$. Substituting $D = S + (T_d - 1)A$ and (15) into (16) gives

$$\sum_{i=1}^g \chi(c_i) \alpha(c_{i,x}) + 2 \sum_r \alpha(r) \equiv 0 \pmod{4}. \quad (17)$$

Choosing $\alpha = \mathbf{1}_{\{r\}}$ (the indicator of one residue) turns (17) into $\sum_{i: c_{i,x} \equiv r} \chi(c_i) + 2 \equiv 0$, i.e. (13), using $-2 \equiv 2 \pmod{4}$. Rotation gives the b -statement. For $a = 0$: take a finitely supported exact-column $\alpha: \mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$; then b is even, w is d -periodic, vertical two-periodicity makes the monodromy zero, and exact-column indicators give the $\equiv 0$ statement; rotation gives exact rows. $\square$

Corollary 7.2 (P16: strand-density bound). *Every finite-support repeating trace with nonzero drift $d = (a, b)$ has*

$$g \geq 2 \max(|a|, |b|). \quad (18)$$

In particular $g = 0$ is impossible: zero-growth repeating traces do not exist (an independent proof of Theorem 6.1). If a residue fiber contains exactly two strands, both bases share checkerboard parity.

Proof. A sum of m signs ± 1 that is $\equiv 2 \pmod{4}$ is nonzero and has m positive and even. Hence each of the $|a|$ residue classes mod $|a|$ (and each of the $|b|$ classes mod $|b|$) contains at least two strand bases, giving $g \geq 2|a|$ and $g \geq 2|b|$. If $g = 0$ some class is empty while $d \neq 0$, contradicting (13). A two-strand fiber has χ -sum ± 2 , forcing equal signs. $\square$

Unlike the even-winding proof, Theorem 7.1 needs neither the axis alignment of Proposition 5.2 nor the clean-packet reduction; it applies directly to every repeating trace.

8. COLLISION-CHAIN PARITY

Encode a finite cell set D as edges of the bipartite row-column incidence graph: cell (x, y) is the edge $C_x R_y$. At a time s , assign the token $\eta_s = C_{p_s, x}$ if q_s is vertical and $\eta_s = R_{p_s, y}$ if q_s is horizontal.

Theorem 8.1 (Two-endpoint collision-chain parity). *Over $\mathbb{F}_2$, the boundary (set of odd-degree vertices) of the row-column incidence graph of $B_s \triangle B_t$ equals $\eta_s + \eta_t$. Intermediate tokens telescope across arbitrary collision checkpoints; hence at most one defect component is non-Eulerian, and a same-pose return ($\eta_s = \eta_t$) has only Eulerian defect components.*

Proof. One step toggles the single edge $C_{p_{s,x}}R_{p_{s,y}}$, changing the degree parity of exactly its two endpoints. If the pre-step heading is vertical then $\eta_s = C_{p_{s,x}}$ and the post-step heading is horizontal with unchanged row, so $\eta_{s+1} = R_{p_{s,y}}$; for a horizontal pre-step heading the roles are reversed. Hence $\eta_s + \eta_{s+1} = C_{p_{s,x}} + R_{p_{s,y}}$, exactly the one-step boundary contribution. Summing over a run telescopes to $\eta_s + \eta_t$. By the handshaking lemma, every connected component has an even number of odd-degree vertices. If $\eta_s \neq \eta_t$, the graph has exactly those two odd vertices, so they lie in the same component, which is the unique non-Eulerian component. If $\eta_s = \eta_t$, the boundary is empty and every component is Eulerian. $\square$

This identity gives a four-cell lower bound on completed lifetimes. Let s be immediately after an R visit to $z = (x, y)$ and t immediately before its next (L) visit. Then z is black at both times, so $C_x R_y$ is absent from $G := B_s \triangle B_t$. If arrivals at z are vertical, the R step leaves z horizontally along row y , so $\eta_s = R_y$, while $\eta_t = C_x$; for horizontal arrivals the roles reverse. Hence $\partial G = C_x + R_y$. The two vertices are joined in G by a path avoiding $C_x R_y$; in the simple bipartite incidence graph such a path has at least three edges. Adjoining $C_x R_y$ closes a cycle of length at least four, so the completed lifetime involves at least four distinct cells.

Circuit-partition and interlacement formalisms may help compress the remaining chronological constraints, but no touch-graph rank theorem is claimed here.

9. TRANSVERSE RIGIDITY AND ALL-PERIOD NARROW-WIDTH EXCLUSIONS

The constraints above are incidence statements. Section 11 records countermodels that satisfy every one of them and are still not realisable, so no sharpening of the residue equations alone can close the classification problem. This section proves statements of a different kind, using the *order* in which a cell is visited and exact local dynamics rather than only aggregate counts. The width-two and computer-assisted width-four exclusions both hold at every period.

Assume the drift is diagonal: $d = (a, b)$ with $|a| = |b| = m \geq 1$. A quarter-turn rotation sends d to $(-b, a)$, hence ab to $-ab$, and commutes with the ant's rule; since highways are identified up to quarter-turn rotation we normalise to $ab < 0$ and set $t(x, y) = x + y$, so $d = me$ with $e = \pm(1, -1)$. Then $t(d) = 0$, so t is constant on translation classes, and every unit step changes t by exactly ± 1 : the increments are $+1$ for east and north, -1 for west and south. (For an axis drift the transverse functional is x or y , which two moves fix, and everything below fails.) Put $T = \max_i t(p_i)$, $T' = \min_i t(p_i)$ and $W = T - T'$. All cells of a level line $t = c$ share the checkerboard parity of c , so by the HV partition each level line is entirely horizontal-arrival or entirely vertical-arrival.

The normalisation is essential rather than cosmetic. In the frame $ab > 0$ one must take $t = x - y$, whose t -increasing headings are east and *south*; there the argument below eliminates the horizontal case and the extremal lines are *vertical*. Only the axis-free conclusions—every extremal turn is R, extremal cells are entered at most once, W is even—read identically in both frames.

Theorem 9.1 (extremal-line rigidity). *For a finite-support periodic highway with diagonal drift, normalised as above, both extremal level lines $t = T$ and $t = T'$ are horizontal; the arrival is east on $t = T$ and west on $t = T'$; every turn on them is R; every extremal cell reached by the periodic tail is entered exactly once during that tail, so every translation class meeting an extremal line is a singleton and its tail cells are left black and never revisited thereafter; and W is even.*

Proof. Let I be a class meeting $t = T$ and let $z_n = b_I + nd$, all on that line since $t(d) = 0$. Take n with z_n reached by every phase of I , so the turn sequence at z_n is the stabilised word of I and every visit has a predecessor. Arrivals at z_n increase t and departures do not. Recall R sends north, east, south, west to east, south, west, north, and L reverses this. Were the line vertical, the arrival would be north, whose clockwise image east also increases t and is forbidden, while its anticlockwise image west is allowed; every turn would be L, so the stabilised word would begin with L, contradicting the criterion. So the line is horizontal with arrival east, whose anticlockwise image north is forbidden

and clockwise image south allowed: every turn is R. At $t = T'$ arrivals decrease t and departures must increase it; the vertical case gives arrival south, clockwise image west forbidden, anticlockwise east allowed, hence all L and the same contradiction, so that line is horizontal with arrival west and every turn R. In both cases the stabilised word is a block of R's and must alternate, so $|I| = 1$. Both extremal lines being horizontal, they share a checkerboard parity, so $T \equiv T' \pmod{2}$. $\square$

Theorem 9.2 (no width-two highway). *No finite-support periodic highway with diagonal drift has $W = 2$; hence W is even and $W \geq 4$.*

Proof. Label the three lines 0, 1, 2 for $t = T, T-1, T-2$; lines 0, 2 are extremal and horizontal, line 1 is vertical. Each step changes t by ± 1 , so the ant alternates between line 1 and $\{0, 2\}$. Orienting the longitudinal axis so that $e = (1, -1)$, the excursions are forced by Theorem 9.1: from a line-1 cell z the upward excursion enters line 0 heading east and leaves heading south, carrying $z \mapsto z + e$; the downward one enters line 2 heading west and leaves heading north, carrying $z \mapsto z - e$. Index line-1 cells by $\ell \in \mathbb{Z}$ along e .

An upward excursion out of ℓ consumes the line-0 cell U_ℓ and a downward one the line-2 cell D_ℓ ; distinct sites consume distinct cells, each entered at most once by Theorem 9.1. The gap $(\ell, \ell + 1)$ is crossed upward only from ℓ and downward only from $\ell + 1$, so it is crossed at most once in each direction. Choose a cyclic phase at a visit to line 1, and let s_0 be that starting site. Drift $+m$ sends the site to $+\infty$, so every gap above s_0 has net crossing number exactly $+1$; with each count at most one this forces one upward and no downward crossing. Hence every site $\ell > s_0$ is entered exactly once, from $\ell - 1$, and departs to $\ell + 1$.

Entering from $\ell - 1$ completes an upward excursion, which leaves line 0 heading south, so the ant arrives heading south; departing to $\ell + 1$ begins an upward excursion, so it leaves heading east. The anticlockwise image of south is east, so the unique turn at ℓ is L. The class of such a site has cells $\ell + nm$, almost all above s_0 and hence entered once, so it is a singleton whose stabilised word is the single letter L—contradicting the criterion. $\square$

We record the two errors this argument passed through, since both are natural traps. Writing the transverse functional with a case split and then arguing in one frame gives a false “both extremal lines are horizontal” in the rotated frame, where they are vertical; and asserting that a site’s first visit turns R is wrong, because the criterion constrains only stabilised words and the seed construction blackens cells below their class’s maximal level precisely so that some first turns are L. The proof above avoids the first by normalising and using explicit compass directions, and the second by counting edge crossings rather than site visits.

The residue theorem and extremal parity also give a useful general inequality. If the normalised drift is $(m, -m)$, growth is g , and transverse width is W , then

$$2m \leq g \leq mW. \quad (19)$$

Indeed a translation class is labelled by $(t, x \bmod m)$. In each fixed x -residue the signed residue identity makes the number of odd classes positive and even. There is at most one on each of the $W + 1$ levels; since W is even, the largest possible even number is W . Summing over the m residues proves the upper bound, and the strand-density theorem gives the lower bound. For the standard highway $(m, g, W) = (2, 12, 10)$ this yields only $W \geq 6$.

The width-four case can nevertheless be excluded directly. Rotate by 180° if needed, take drift $(m, -m)$ with $m > 0$, shift the transverse range to $t = x + y \in \{0, 1, 2, 3, 4\}$, and view each fixed- x column as a five-bit tape symbol. At a half-integer boundary record, in chronological order, the transverse line of the destination cell after each horizontal crossing. Directions alternate and are implicit: rightward entries are 2 or 4, leftward entries are 0 or 2. Extremal rigidity permits rightward 4 and leftward 0 at most once on each boundary.

Lemma 9.3 (exact blank-column graph). *For one initially white five-cell column, among boundary sequences that alternate right/left beginning and ending right, use only rightward labels in $\{2, 4\}$ and*

leftward labels in $\{0, 2\}$, and use rightward 4 and leftward 0 at most once on each boundary, the complete list of compatible left-to-right pairs is

202	4
4220222	2, 202, 20222, 2022224
2042222	2, 202, 20222, 2022224
202242222	422, 42202, 4220222.

This twelve-edge graph is acyclic and has longest directed path three.

Exact computer-assisted verification. Literal simulation tracks the 32 colour masks, current entry line and heading, and four flags for the one allowed extreme event on either side. The complete recursion reaches 22 entry states, has no reachable state cycle, and terminates by depth eight. A second implementation, using compass headings and sets of black rows and sharing no classifier code, gives the same twelve pairs. It also extracts two far cuts from the documented standard-highway seed and exactly replays the initially white intervening column. Assign ranks 3, 2, 1, 0 respectively to 202242222, the pair 4220222, 2042222, the vertex 202, and all other displayed vertices. Every edge lowers rank; this last finite implication is checked in Lean. The Lean file does not formalise completeness of the two Python enumerations. $\square$

Theorem 9.4 (no width-four highway). *No finite-support periodic highway with diagonal drift has transverse width $W = 4$. Hence every such highway has even width $W \geq 6$.*

Proof. At the beginning of the periodic tail only finitely many cells have been visited and only finitely many cells are black, by finite initial support. Hence all sufficiently far columns are untouched and white. At any far boundary the longitudinal coordinate is eventually nm plus a bounded phase offset; its complete crossing sequence is therefore finite, nonempty, and alternates right, left, ..., right. For every far column, its left and right sequences form an edge of Lemma 9.3. Infinitely many consecutive white columns would give an infinite directed path, contradicting the rank bound. $\square$

There is an independent arithmetic view of the width-four geometry.

Proposition 9.5 (width-four masks and period bound). *Let a finite-support periodic highway have transverse width four and normalised drift $(m, -m)$, $m > 0$. For each longitudinal residue, the set of levels carrying odd translation classes is one of*

$$\{0, 2\}, \{1, 3\}, \{0, 4\}, \{2, 4\}, \{0, 1, 2, 4\}, \{0, 2, 3, 4\};$$

and for each diagonal coordinate residue the corresponding set belongs to the same six-element list. If h of the m longitudinal residues use a four-element mask, then $g = 2m + 2h$ and $h \equiv m \pmod{2}$. If P is the macro period and $N = 2P$ the ant period, then

$$P \leq 7m, \quad N \leq 14m, \quad P \equiv m \pmod{2}.$$

Proof. Let $o_{j,r}$ indicate that the class on level j and longitudinal residue r is odd. The signed residue identity is $o_{0,r} - o_{1,r} + o_{2,r} - o_{3,r} + o_{4,r} = 2 \pmod{4}$. Exhausting five binary values gives exactly the six displayed masks; applying the other coordinate residue identity gives the diagonal version. Summing mask sizes gives $g = 2m + 2h$, and $g \in 4\mathbb{Z}$ gives the parity of h .

For the period bound, cyclically start immediately after a bottom-extremal visit and observe the ant only on the odd levels. The forced bottom turn puts it in state LN , and one translated period returns to LN ; width four ensures that a top-line transition also occurs. Write the four states LN, LS, UN, US for lower/upper level and north/south arrival. The twelve possible two-turn macro transitions are

LN	$(LS, +1, RR)$	$(UN, +1, RL)$	$(LN, -1, LR)$
LS	$(LN, -1, RR)$	$(LS, +1, LR)$	$(UN, +1, LL)$
UN	$(US, +1, RR)$	$(UN, -1, LR)$	$(LS, -1, LL)$
US	$(UN, -1, RR)$	$(LS, -1, RL)$	$(US, +1, LR)$

Let x count $LN \rightarrow LN$, let y count $US \rightarrow US$, let c count $LS \rightarrow LN$, and let f count $UN \rightarrow US$. If a, b count $LN \rightarrow LS, LN \rightarrow UN$ and i, j count $US \rightarrow UN, US \rightarrow LS$, flow conservation gives $c = a + b$ and $f = i + j$. Extremal singleton rigidity gives $x + c \leq m$ and $f + y \leq m$. If L_1, L_3 are the lower and upper odd-level left-turn totals, the P3 condition that each stabilised class alternates and begins with R makes the total $R - L$ excess on each odd level nonnegative. The lower right-turn total is $a + b + c = 2c$ and the upper one is $f + i + j = 2f$, so $L_1 \leq 2c$ and $L_3 \leq 2f$. Flow conservation and the displacement read from the table give

$$P = 2c + 2f + L_1 + L_3, \quad m = L_1 - L_3 - 2x + 2y.$$

Thus $L_1 + L_3 \leq 3m - 6x + 2y$, and substitution yields $P \leq 7m - 8x \leq 7m$. Every macro move changes the longitudinal coordinate by ± 1 , so $P \equiv m \pmod{2}$. $\square$

Combined exact searches cover the entire theorem-bounded range for $1 \leq m \leq 9$: the fixed-drift engine supplies the full range for $m \leq 6$ and the upper ranges for $m = 7, 8, 9$, while the general period- ≤ 48 and unrestricted width-four searches supply their lower ranges. The result is zero hits throughout. The $m = 10$ run is partial through $P = 66$, and its unfinished rows are not counted. These arithmetic and search results are cross-checks and structural data; Theorem 9.4 does not depend on them or on the residue theorem.

The proof has a sharp width boundary. At widths six and eight the local recursion contains a reachable two-state oscillator: black rows $\{1, 2\}$ with an eastward entry on row 2 change to $\{1, 3\}$ and exit eastward, and a westward return on row 2 restores the first state. The reused cells alternate their turns and neither extreme repeats. This is not a highway, but it shows that one-column acyclicity does not extend unchanged. Nor does a finite column alphabet alone make an unbounded tape finite-state; wider strips require a genuine bounded-memory theorem.

10. EXACT COMPUTATIONAL THEOREMS

All computations use exact integer coordinates, exact finite sets, and no probabilistic equality tests. The residue-free Java variant certifies the period-34–48 exclusion. Its strand- and residue-pruned variants share one enumeration framework and are consistency checks, not independent implementations. A separately written Python verifier independently reproduces the complete exclusion through length 18 and audits the exact criterion.

Theorem 10.1 (gateway seed). *The blank orbit first enters its repeating trace at step 9977; the trace has least period 104 and normalised drift $(-2, 2)$. Applying Theorem 3.1 to this word yields the 13-cell black seed*

$$\begin{aligned} &(-6, 0), (-5, 0), (-5, 1), (-4, 0), (-4, 2), (-3, -1), (-3, 2), \\ &(-2, -1), (-2, 1), (-1, 1), (0, 0), (0, 1), (0, 2), \end{aligned}$$

which, with a north-facing ant at the origin, produces the 104-step highway immediately; the semi-infinite leading corridor is white and the gateway induction recurs translated by the drift.

Theorem 10.2 (bounded-period exclusion). *No finite-support periodic highway of nonzero drift has period ≤ 48 . The verification proceeds by:*

- (a) *a complete first-R, cyclic-phase-normalised enumeration through period 32 (46,185,421 nodes at depths 6–32 over 32 prefix shards, plus a 59-node unprefix baseline through depth 5; 46,185,480 nodes in total). Every positive-growth heading-resetting word contains R, so a cyclic shift places it in this enumeration; shifting changes only the phase on the same periodic orbit and preserves finite-support realisability;*
- (b) *rank-by-rank audited positive-growth searches at periods 34, ..., 48 with the residue-free variant, which uses no consequence of Theorem 7.1 and applies the exact criterion to every positive-growth nonzero-drift leaf, using Corollary 6.2 to skip zero growth; odd periods restore no heading. The exact node and leaf counts are:*

<i>Period</i>	<i>Nodes</i>	<i>Leaves</i>
34	40,418,033	6,989,418
36	108,995,287	18,928,333
38	293,763,689	51,386,767
40	791,801,484	139,000,900
42	2,133,302,151	376,781,423
44	5,748,138,964	1,018,293,035
46	15,483,269,352	2,757,152,898
48	41,710,394,384	7,446,719,550

At period 48 all 2^{16} prefix ranks were covered with no node cap. The exact criterion was evaluated at every leaf and returned zero certificates at every period; every rank was covered exactly once. Strand-pruned and residue-pruned variants, which do assume Theorem 7.1/7.2, return zero certificates on the strictly smaller trees they explore and serve as cross-checks only.

Together with Corollary 6.2 (which excludes zero growth at every period) this excludes every finite-support periodic highway of period at most 48.

These theorems are exact for the stated finite ranges only. They bound the least period of a possible nonstandard highway, whereas Conjecture 1.1 quantifies over every finite configuration and every period; this finite search does not exhaust that range.

11. REMAINING GAPS AND OPEN PROBLEMS

The results above leave two logically separate problems. The first is a universal entrance problem; the following are two sufficient formulations, which are not claimed to be equivalent.

Problem 11.1 (bounded-retreat lemma). For every finite-support initial configuration there exist a diagonal direction, a time T , and a width $W < \infty$ such that after T the ant never retreats more than W layers behind its current record frontier and all transverse excursions stay in a width- W corridor.

Problem 11.2 (bounded active core). For every finite-support initial configuration there are a finite window $W \subset \mathbb{Z}^2$, a time T , and translations v_t such that, for every $t \geq T$, the ant and every cell or Tait-boundary component capable of influencing the future orbit lie in $v_t + W$; all components outside that translated window are permanently inert and never reabsorbed.

Under either formulation, the colours in a bounded frontier window, the ant's position and heading form a finite state modulo translation, so determinism forces eventual periodicity. This would not identify the resulting cycle with the standard highway; that is the separate periodic-classification problem below. No proof of either entrance formulation is known, and the following obstructions show why:

- Unboundedness (Theorem 2.4) and fixed-domain escape do not imply bounded retreat: an orbit may keep enlarging a two-dimensional region while escaping every fixed box.
- No nonconstant affine combination of (E, V, k, β) —in particular no single scalar Tait invariant among them—is monotone (Proposition 4.4).
- A local pumping argument is invalid: equal local cores can be joined through different distant connectivity, changing a later merge into a split (Theorem 4.1). A valid finite-state proof must first establish permanent separation from the discarded graph or a finite connectivity summary.
- At fixed growth $g = 4$ the signed-residue skeletons of Theorem 7.1 leave unbounded free parameters; six exact heading-reset countermodels satisfy every incidence/residue condition yet fail the cross-period criterion, so these displayed incidence and residue conditions alone cannot finish the periodic branch.

Problem 11.3 (periodic classification). Prove that every positive-growth periodic highway admitted by finite support agrees in path and turn trace with the standard period-104 highway up to lattice translation, quarter-turn rotation, and cyclic phase shift. One identified route requires a single-tour chronology theorem valid for unbounded period; the bounded-period exclusions of Section 10 are only a finite frontier. Theorem 9.4 removes one unbounded family, but widths six and above, non-diagonal translators, and uniqueness within the surviving cases remain.

A genuine *disproof* would output an infinite certificate: either a nonstandard periodic word passing Theorem 3.1 (disproving uniqueness of the highway), or a finite self-enlarging Tait gadget whose state recurs with a parameter increased (an induction proof of nonperiodic growth). No such certificate has been found.

12. SUMMARY

We have assembled a coherent theory of the finite-support Langton ant: an exact decision criterion for periodic highways, a Tait-graph conjugacy with surgery calculus, two independent proofs that no zero-growth clean translator exists (hence every highway grows a wake of size $\in 4\mathbb{Z}_{>0}$), the strand-density bound $g \geq 2\|d\|_\infty$, exhaustive exclusion of all periodic highways of period ≤ 48 , extremal-line rigidity, and all-period exclusions of diagonal transverse widths two and four. The width-four theorem is computer-assisted, with two independent local enumerators and a Lean-checked graph rank certificate. We also give reductions relevant to two remaining problems: universal entrance and all-period classification. Selected algebraic kernels are machine-checked in Lean 4; the geometric extraction is not formalized end to end (Appendix A). Conjecture 1.1 remains open; neither remaining problem is solved here.

APPENDIX A. LEAN 4 FORMALIZATION OF SELECTED ALGEBRAIC KERNELS

A dependency-light Lean 4 project (pinned to Lean 4.32.0, standard library only) machine-checks the exact transition kernel and the algebraic invariants used above. All audited theorems depend only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`; there are no `sorry`s, no custom axioms, and no `native_decide`. The checked statements include:

- the finite-support transition kernel and its toggle lemmas (the departed cell flips; every other cell is unchanged);
- the heading/black-count reset theorem (Proposition 2.5, mod4);
- the finite parity telescope `evenWindingParityCore`: adjacent half-flow parities bookended by zero exterior values have zero aggregate seam parity—the core (11) of Theorem 6.1—and the contradiction wrapper `noOddWindingFromFiniteHalfFlow`;
- the four-corner admissibility and mod-four fiber arithmetic of Theorem 7.1: $E_r + 2 = 0 \Rightarrow E_r = 2$ in $\mathbb{Z}/4$, a residue fiber = 2 has a positive even number of bases, residue fibers partition the base list, and one identity per residue implies $g \geq 2n$;
- finite potential telescoping, checker-cycle monodromy $2\sum \alpha$, and translated-widget cancellation (the steps of (15)–(17));
- collision-chain endpoint parity (Theorem 8.1) and the separately formalized exact directed-pose row/column $\mathbb{Z}/4$ discrepancy charges.
- the width-four arithmetic kernels (six-mask enumeration, period–drift inequality, period parity, and the primitive arithmetic reduction), and the finite rank certificate for Lemma 9.3. Lean checks every listed graph edge, not completeness of the independently generated edge table.

The formalization deliberately isolates, as an explicit interface, the two inputs it does not derive from the kernel: the endpoint normal form and the lifted potential geometry of a genuine periodic ant trace certificate. Consequently neither Theorem 7.1/7.2 nor Conjecture 1.1 is claimed as an unconditional Lean theorem; what is certified is the algebraic layer conditional on a structured trace certificate, together with the growth bound it yields.

REPRODUCIBILITY

The exact search engines, the periodic-trace checker, the Lean project, and all period-exclusion records with SHA-256 manifests accompany this work in the supplementary repository [7]. The standard word, the 13-cell seed, and every period- ≤ 48 shard are regenerable from the sources. The period-34–48 certifying shards record `search_complete` and `node_cap=null`. The period- ≤ 32 prefix shards record a 10^8 safety cap, which every shard left unreached (the largest used 2,623,020 nodes). Released audit scripts verify rank coverage, aggregate counters, and zero-hit outcomes; agreement among the three Java variants is a consistency check rather than independent replication. The width-four supplement additionally contains the two independent blank-column classifiers, their twelve-edge JSON certificate, the Lean rank module, the complete fixed-drift summary for $m \leq 9$, and the explicitly partial $m = 10$ record.

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